When can $\sum f(n)/n$ vanish for a periodic f with values ± 1?

Contracting 2-adic digit trees, a sharp obstruction for periods $2^a p$, and the Murty–Pathak problem for periods $N \equiv 0 \pmod 4$

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Abstract

Let $\mathcal{D}$ be the set of periods N for which some N-periodic $f\colon \mathbb{Z} \to \{\pm 1\}$ has $\sum_{n \le N} f(n) = 0$ and $\sum_{n \ge 1} f(n)/n = 0$. Murty and Pathak proved that $N \notin \mathcal{D}$ for $N \equiv 2 \pmod 4$, recorded Tengely's example $36 \in \mathcal{D}$, and left the periods $N \equiv 0 \pmod 4$ open; a recent manuscript asked whether a *non-square* $N \equiv 0 \pmod 4$ can lie in $\mathcal{D}$. We introduce a 2-adic method for Okada's criterion. The conditions attached to characters of 2-power conductor are equivalent to one functional equation on binary trees of signs ("digit trees"), $G(u) + pZ(u) - Z(pu) = K$ on $(\mathbb{Z}/2^a\mathbb{Z})^\times$, and the Euler factor $p - T_p$ acts on differences of sibling subtrees as a contraction. We prove: (A) $2^a \notin \mathcal{D}$ for every a; (B) if p is an odd prime and $2^{a-1} \le p - \chi_4(p)$, then $2^a p \notin \mathcal{D}$; in particular $4p, 8p \notin \mathcal{D}$ for all odd primes and, for fixed a, only finitely many $2^a p$ lie in $\mathcal{D}$. Conversely, (E) an explicit digit-tree criterion produces members of $\mathcal{D}$, and (C) the threshold $a_0(p) = \min\{a : 2^a p \in \mathcal{D}\}$ equals 4 for $p = 3$ and $2^k + 2$ for every Fermat prime $p = 2^{2^k} + 1$ with $k \ge 1$, so that (B) is sharp. Finally (D) three p-independent Laurent-polynomial certificates show that $12p \in \mathcal{D}$ for every prime $p \ge 5$ with $p \not\equiv \pm 1 \pmod{24}$, and each such $12p$ is a minimal element of $\mathcal{D}$. Consequently the answer to the non-square question is negative, 48 is the least non-square element of $\mathcal{D}$ and 36 the least element, both lower bounds now without computer search, and $\mathcal{D}$ has infinitely many minimal elements. In the range of (A) and (B) every convergent sum $\sum f(n)/n$ is transcendental.

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1 Introduction

1.1 The problem

For a periodic arithmetical function f with period N the series $L(1,f)=\sum_{n\ge1} f(n)/n$ converges if and only if $\sum_{n=1}^{N} f(n)=0$. Chowla [7] asked when $L(1,f)$ can vanish; Baker, Birch and Wirsing [2] answered this for prime periods, and Okada [15] turned the vanishing of $L(1,f)$ for rational-valued f into an explicit system of $\varphi(N)+\omega(N)$ linear equations. By Baker's theorem on linear forms in logarithms, $L(1,f)$ is then either 0 or transcendental [1, 2, 13].

Two sign problems have grown out of this. Erdős conjectured (see Livingston [11]) that $L(1,f)\neq0$ when $f(n)=\pm1$ for $q\nmid n$ and $f(n)=0$ for $q\mid n$; this was proved for $2\varphi(q)+1>q$ by Okada, for $q\equiv3 \pmod 4$ by Murty and Saradha [14], for a set of $q\equiv1 \pmod 4$ of lower density at least 0.82 by Chatterjee and Murty [5], "with probability one" by Pathak [16], and for further families by Bharadwaj [3]. Two recent manuscripts show that the conjecture is false, the least counterexample being $q=105^2$ [8], and that it holds for every non-square modulus [10].

The second problem, which is the subject of this paper, concerns functions with $f(n)=\pm1$ for *all* n.

Definition 1.1. $\mathcal{D}$ is the set of integers $N\ge1$ for which there is an N-periodic $f\colon\mathbb{Z}\to\{-1,1\}$ with $\sum_{n=1}^{N} f(n)=0$ and $\sum_{n\ge1} f(n)/n=0$.

Clearly $kN\in\mathcal{D}$ whenever $N\in\mathcal{D}$, and $N\in\mathcal{D}$ forces N to be even. Murty and Pathak [12] proved that $N\notin\mathcal{D}$ for $N\equiv2 \pmod 4$, reported Tengely's example showing $36\in\mathcal{D}$ (the least possible period), and left the case $N\equiv0 \pmod 4$ open [12]; see also [10, Sect. 6]. Since $36=6^2$ and since the companion problem for Erdős functions turned out to be governed by squares [8, 10], the following was asked in [10, Sect. 6].

Problem 1.2 ([10, Sect. 6])**.** Let $N\equiv0 \pmod 4$ be a non-square and let f be N-periodic with $f(n)=\pm1$ for all n and $\sum_{n=1}^{N} f(n)=0$. Is $\sum_{n\ge1} f(n)/n\neq0$?

The answer is *no*, and the reasons are more interesting than a single example: whether $N\in\mathcal{D}$ is decided by a competition between the 2-adic depth of N and the size of its odd prime factors. We make this precise through the following new problem.

Problem 1.3 (threshold problem)**.** For an odd prime p let $a_0(p)=\min\{a\ge1:\ 2^a p\in\mathcal{D}\}\in\{1,2,\dots\}\cup\{\infty\}$. Determine $a_0(p)$.

Since $\mathcal{D}$ is closed under multiples, $2^a p\in\mathcal{D}$ if and only if $a\ge a_0(p)$.

1.2 Results

Write χ_4 for the non-trivial character modulo 4.

Theorem A. *$2^a \notin \mathcal{D}$ for every $a \geq 0$.*

Theorem A is the simplest instance of the method; we have not found it stated in the literature.

Theorem B. *Let p be an odd prime and $a \geq 1$ with $2^{a-1} \leq p - \chi_4(p)$. Then $2^a p \notin \mathcal{D}$. Equivalently, $a_0(p) \geq \lfloor \log_2(p - \chi_4(p)) \rfloor + 2$.*

Thus $4p \notin \mathcal{D}$ and $8p \notin \mathcal{D}$ for every odd prime p, $16p \notin \mathcal{D}$ for $p \geq 7$, $32p \notin \mathcal{D}$ for $p \geq 17$, and for every fixed a only finitely many primes p have $2^a p \in \mathcal{D}$. The case $a = 1$ is the case $N = 2p$ of the theorem of Murty and Pathak.

For the converse direction we give a sufficient criterion (Theorem E in Section 6) in terms of an odd, 2-adically 1-Lipschitz function H' on $(\mathbb{Z}/2^J\mathbb{Z})^\times$; it produces explicit functions f and shows that Theorem B is sharp for $p = 3$ and for every Fermat prime.

Theorem C. *$a_0(3) = 4$, and $a_0(p) = 2^k + 2$ for every Fermat prime $p = 2^{2^k} + 1$, $k \geq 1$. In particular $a_0(5) = 4$, $a_0(17) = 6$, $a_0(257) = 10$ and $a_0(65537) = 18$, and for these primes $2^a p \in \mathcal{D}$ if and only if $2^{a-1} > p - \chi_4(p)$.*

Complete computer searches combined with Theorem E give in addition $a_0(7) = a_0(11) = a_0(13) = 7$, and Theorem E gives $a_0(p) < \infty$ for every prime $p < 127$ (Table 2 and Section 10).

The third family of periods is $12p$.

Theorem D. *For every prime $p \geq 5$ with $p \not\equiv \pm 1 \pmod{24}$ we have $12p \in \mathcal{D}$, and $12p$ is a minimal element of $\mathcal{D}$ with respect to divisibility.*

The proof of Theorem D consists of three certificates, one for each pair of Legendre symbols $(\frac{2}{p}), (\frac{3}{p})$ other than $(1, 1)$; each certificate is an identity between Laurent polynomials in two variables and is independent of p (Table 1). For the 10 primes $p < 250$ with $p \equiv \pm 1 \pmod{24}$ we found f with $12p \in \mathcal{D}$ by computer (Section 9).

Corollary 1.4. (i) *The answer to Problem 1.2 is negative. The least non-square element of $\mathcal{D}$ is 48, and every $2^a p$ with $p \in \{3, 5\}$, $a \geq 4$, every $2^a p$ with $p \geq 5$ a Fermat prime and $2^{a-1} > p - 1$, and every $12p$ as in Theorem D is a non-square element of $\mathcal{D}$.*

(ii) *The least element of $\mathcal{D}$ is 36.*

(iii) *$\mathcal{D}$ has infinitely many minimal elements; in particular $\mathcal{D}$ is not the set of multiples of finitely many integers.*

(iv) *If $N = 2^a$, or $N = 2^a p$ with $2^{a-1} \leq p - \chi_4(p)$, then $\sum_{n \geq 1} f(n)/n$ is transcendental for every N-periodic $f \colon \mathbb{Z} \to \{\pm 1\}$ with $\sum_{n \leq N} f(n) = 0$.*

The lower bounds in (i) and (ii) follow from Theorems A and B and the theorem of Murty and Pathak; they replace a computer search. The examples for 36 and 48 are listed in Appendix A.

1.3 The method

Okada's criterion splits into conditions indexed by primitive characters. For $N = 2^a p$ we group the residues according to their 2-adic level and read the values of f on the residues above a unit $u \in (\mathbb{Z}/2^a\mathbb{Z})^\times$ as the binary digits of an odd integer (Lemma 3.2). The conditions for all characters of conductor 2^j and $2^j p$ ($2 \leq j \leq a$) then collapse, by a telescoping identity (Lemma 3.3), into a single *master equation*

$$\widehat{G}(u) + p\,\widehat{\Phi}_y(u) - \widehat{\Phi}_y(pu) = K + (p-1)\beta(y) \qquad (u \in (\mathbb{Z}/2^a\mathbb{Z})^\times,\ y \in (\mathbb{Z}/p\mathbb{Z})^\times)$$

between the value functions of the digit trees of f (Proposition 5.1). Differences of sibling subtrees satisfy $\delta(pu) = p\,\delta(u) + \gamma(u)$, and since multiplication by p permutes the units, a maximum argument (Lemma 3.4) gives $|\delta| \le |\gamma|/(p-1)$: the Euler factor contracts. When the depth is small compared with p, i.e. $2^{a-1} \le p - \chi_4(p)$, the contraction kills all 2-adic structure, and the problem descends to the residues modulo p, where a second, multiplicative contraction (by the factor $2 - T_2$ coming from the prime 2) and a parity count finish the proof. The same digit trees, run in reverse, produce the examples.

1.4 Organisation

Section 2 recalls Okada's criterion. Section 3 contains the lemmas on digit trees. Theorem A is proved in Section 4, Theorem B in Section 5, Theorems E and C in Section 6, Theorem D in Section 7, and the corollaries in Section 8. Section 9 describes the computations (none of which is used in the proofs of Theorems A–E), and Section 10 lists open problems.

2 Okada's criterion

Let $N \ge 2$. For $d \mid N$ put $U_d = (\mathbb{Z}/d\mathbb{Z})^\times$, with $U_1 = \mathbb{Z}/1\mathbb{Z}$ a single point. The residues n modulo N with $\gcd(n, N) = N/d$ are the $n = (N/d)x$, $x \in U_d$; we call them the *layer* of level d (the layer $d = 1$ is the residue 0). For an N-periodic f put

$$f_d(x) = f\big((N/d)x\big)\ (x \in U_d), \qquad S_d = \sum_{x \in U_d} f_d(x), \qquad \langle f_d, \chi \rangle = \sum_{x \in U_d} f_d(x)\overline{\chi(x)}$$

for Dirichlet characters χ whose conductor divides d (regarded as characters modulo d), and for a prime $r \mid N$

$$\varepsilon_r(d) = v_r(N/d) - \frac{[\,r \mid d\,]}{r-1}.$$

Here $[\,P\,]$ is 1 if P holds and 0 otherwise.

Theorem 2.1 (Okada [15]; layered form [3, Thm. 4], [8, Thm. 2.2], [10, Thm. 2.1 and Sect. 6])**.** *Let $f\colon \mathbb{Z} \to \mathbb{Q}$ be N-periodic with*

$$\text{(C)} \qquad \sum_{n=1}^{N} f(n) = \sum_{d \mid N} S_d = 0.$$

Then $\sum_{n \ge 1} f(n)/n = 0$ if and only if

(A) *for every non-principal primitive character χ whose conductor m divides N,*

$$\sum_{d:\ m \mid d \mid N} \frac{d}{\varphi(d)} \prod_{\substack{r \mid d \\ r \nmid m}} \Big(1 - \frac{\chi(r)}{r}\Big) \langle f_d, \chi \rangle = 0$$

(r runs over primes), and

(B) *for every prime $r \mid N$, $\sum_{d \mid N} S_d\, \varepsilon_r(d) = 0$.*

The sufficiency of (A) and (B) is elementary: grouping $\sum f(n) n^{-s}$ by $\gcd(n, N)$, expanding each f_d in characters modulo d and letting $s \to 1^+$ gives

$$\sum_{n \ge 1} \frac{f(n)}{n} = \frac{1}{N} \sum_{\chi} c_\chi(f)\, L(1, \chi) - \frac{1}{N} \sum_{r \mid N} \Big(\sum_{d \mid N} S_d\, \varepsilon_r(d) \Big) \log r, \tag{1}$$

where χ runs over the non-principal primitive characters of conductor dividing N and $c_\chi(f)$ is the left-hand side of (A). Indeed, the principal characters contribute $\zeta(s)\sum_d (N/d)^{-s}\frac{S_d}{\varphi(d)}\prod_{r\mid d}(1-r^{-s})$, whose pole at $s=1$ has residue $\frac{1}{N}\sum_d S_d = 0$ by (C), and the logarithmic terms come from differentiating $(N/d)^{-s}\prod_{r\mid d}(1-r^{-s})$ at $s=1$. The layer $d=1$ (the residue 0, with $S_1 = f(0)$ and $\varepsilon_r(1) = v_r(N)$) contributes only to (B) and (C); this is the form needed for functions with $f(0)\neq 0$ [10, Sect. 6]. The necessity of (A) and (B) is Okada's theorem [15]; its proof combines Baker's theorem on linear forms in logarithms with a Galois-conjugation argument (see also [2, 3, 8]). We use necessity in Sections 4 and 5 and only sufficiency in Sections 6 and 7.

Lemma 2.2. (a) *If $N\in\mathcal{D}$ then $kN\in\mathcal{D}$ for every $k\geq 1$.*

(b) *If $N\in\mathcal{D}$ then $N\equiv 0 \pmod 4$.*

(c) *If f is N-periodic, ±1-valued, $\sum_{n\leq N} f(n)=0$ and $L(1,f)\neq 0$, then $L(1,f)$ is transcendental.*

Proof. (a) An N-periodic function is kN-periodic. (b) $\sum_{n\leq N} f(n)$ is a sum of N odd numbers, so N is even, and $N\not\equiv 2 \pmod 4$ by [12]. (c) By (1), $L(1,f)$ is a linear form with algebraic coefficients in logarithms of algebraic numbers and in π (the values $L(1,\chi)$ of odd characters are algebraic multiples of $\pi = -i\log(-1)$), and Baker's theorem shows that such a form is 0 or transcendental [1, 2, 13]. □

3 Digit trees

Fix $a\geq 1$. For $0\leq i\leq a$ write $U_{(i)} = U_{2^i} = (\mathbb{Z}/2^i\mathbb{Z})^\times$, so that $U_{(0)}$ and $U_{(1)}$ are single points, and put $u_i = |U_{(i)}|$, i.e. $u_0 = u_1 = 1$ and $u_i = 2^{i-1}$ for $i\geq 1$. For $u\in U_{(a)}$ and $i\leq a$ we write $u \bmod 2^i$ for the image of u in $U_{(i)}$. Put

$$e_i = a - i - [\,i\geq 1\,] \qquad (0\leq i\leq a),$$

so that $e_i = \varepsilon_2(2^i)$ for $N=2^a$ and also $e_i = \varepsilon_2(2^i) = \varepsilon_2(2^i p)$ for $N = 2^a p$.

Definition 3.1. A *digit tree* of depth a is a family $\mathbf{d} = (d_i)_{0\leq i\leq a}$ of functions $d_i\colon U_{(i)}\to\{\pm1\}$. Its *value function* is

$$X_{\mathbf{d}}(u) = \sum_{i=0}^{a} 2^i\, d_i(u \bmod 2^i) \qquad (u\in U_{(a)}).$$

Its *level sums* are $S_i(\mathbf{d}) = \sum_{x\in U_{(i)}} d_i(x)$.

Lemma 3.2 (signed binary expansions)**.** (a) *Every odd integer x with $|x|<2^{a+1}$ has a unique expansion $x = \sum_{i=0}^{a} 2^i d_i(x)$ with $d_i(x)\in\{\pm1\}$, and every such sum is odd of absolute value $<2^{a+1}$.*

(b) *For $0\leq k\leq a$ one has $d_i(x) = d_i(x')$ for all $i\leq k$ if and only if $x\equiv x' \pmod{2^{k+2}}$.*

(c) *A function $X\colon U_{(a)}\to\mathbb{Z}$ is the value function of a digit tree of depth a if and only if all $X(u)$ are odd, $|X(u)|<2^{a+1}$, and $X(u)\equiv X(u') \pmod{2^{k+2}}$ whenever $1\leq k\leq a$ and $u\equiv u' \pmod{2^k}$. The tree is then unique, and $d_0 = -\chi_4(X(u))$ for every u.*

Proof. (a) The map $x\mapsto y = (x+2^{a+1}-1)/2$ is a bijection from the odd integers in $(-2^{a+1}, 2^{a+1})$ onto $[0, 2^{a+1}-1]$, and if $y = \sum_i b_i 2^i$ is the binary expansion then $x = \sum_i 2^i(2b_i - 1)$. (b) The digits $0,\dots,k$ agree iff $y\equiv y' \pmod{2^{k+1}}$ iff $x\equiv x' \pmod{2^{k+2}}$. (c) If $u\equiv u' \pmod{2^k}$ then $u \bmod 2^i = u' \bmod 2^i$ for $i\leq k$, so the digits $0,\dots,k$ of $X(u)$ and $X(u')$ agree and (b) gives the congruence; for $k=1$ this covers the levels 0 and 1, which consist of one point each. Conversely, under the stated conditions put $d_i(x) = d_i(X(u))$ for any $u\in U_{(a)}$ above $x\in U_{(i)}$; by (b) (with $k = \max(i,1)$) this does not depend on u. Finally $x\equiv d_0 + 2d_1 \equiv d_0 + 2 \pmod 4$, so $d_0 = -\chi_4(x)$. □

Lemma 3.3 (telescoping). *Let $A_i\colon U_{(i)} \to \mathbb{C}$ $(0 \le i \le a)$. The following are equivalent:*

(i) *for every j with $2 \le j \le a$ and every primitive character ξ modulo 2^j, $\displaystyle\sum_{i=j}^{a} \sum_{x \in U_{(i)}} A_i(x)\overline{\xi(x)} = 0$;*

(ii) *the function $u \mapsto \sum_{i=0}^{a} 2^i A_i(u \bmod 2^i)$ is constant on $U_{(a)}$.*

Proof. For $1 \le j \le a$ and $x \in U_{(j)}$ let $\Psi_j(x) = \sum_{i=j}^{a} \sum_{x' \in U_{(i)},\ x' \equiv x\ (2^j)} A_i(x')$, so that the sum in (i) is $\sum_{x \in U_{(j)}} \Psi_j(x)\overline{\xi(x)}$. For $j \ge 2$ the non-primitive characters modulo 2^j are exactly those that are trivial on the kernel $\{1, 1+2^{j-1}\}$ of $U_{(j)} \to U_{(j-1)}$. A function on a finite abelian group is invariant under a subgroup iff its Fourier coefficients vanish at all characters that are non-trivial on the subgroup; hence condition (i) for a given j says that $\Psi_j(x + 2^{j-1}) = \Psi_j(x)$ for all $x \in U_{(j)}$ (note $x(1+2^{j-1}) \equiv x + 2^{j-1} \pmod{2^j}$ for odd x).

(i)⇒(ii). For $j < a$ and $x \in U_{(j)}$ we have $\Psi_j(x) = A_j(x) + \Psi_{j+1}(\tilde{x}) + \Psi_{j+1}(\tilde{x} + 2^j)$ for a lift $\tilde{x} \in U_{(j+1)}$, and by the invariance of Ψ_{j+1} this is $A_j(x) + 2\Psi_{j+1}(\tilde{x})$ for either lift. Descending induction from $j = a$ gives $\sum_{i=j}^{a} 2^i A_i(u \bmod 2^i) = 2^j \Psi_j(u \bmod 2^j)$ for $2 \le j \le a$. With $j = 2$ the function in (ii) equals $A_0 + 2A_1 + 4\Psi_2(u \bmod 4)$, which is constant because $\Psi_2(1) = \Psi_2(3)$.

(ii)⇒(i). Let K be the constant. For $2 \le j \le a$ and $x \in U_{(j)}$, each $x' \in U_{(i)}$ $(i \ge j)$ has 2^{a-i} preimages in $U_{(a)}$, so

$$2^a \Psi_j(x) = \sum_{\substack{u \in U_{(a)} \\ u \equiv x\ (2^j)}} \sum_{i=j}^{a} 2^i A_i(u \bmod 2^i) = 2^{a-j}\Big(K - \sum_{i<j} 2^i A_i(x \bmod 2^i)\Big),$$

which depends only on $x \bmod 2^{j-1}$. Hence Ψ_j is invariant under $x \mapsto x + 2^{j-1}$. □

Lemma 3.4 (orbit lemma). *Let π be a permutation of a finite set S, let $q > 1$, and let $\delta, \gamma\colon S \to \mathbb{C}$ satisfy $\delta(\pi(s)) = q\,\delta(s) + \gamma(s)$ for all $s \in S$. Then $\max_S |\delta| \le \max_S |\gamma|/(q-1)$.*

Proof. Choose s with $|\delta(s)|$ maximal. Then $|\delta(s)| \ge |\delta(\pi(s))| \ge q|\delta(s)| - |\gamma(s)|$. □

Lemma 3.5 (weights). (a) *Let $c \in \{-1, 0, 1\}^{a+1}$, $c \ne 0$, with $\sum_{i=0}^{a} u_i c_i = 0$. Then the support of c is $\{0, 1, \dots, k\}$ for some $k \ge 1$, and $c_i = -c_k$ for all $i < k$.*

(b) *For $1 \le k \le a$: $\sum_{i<k} e_i u_i - e_k u_k = 2^k$.*

Proof. (a) Let k be the largest index with $c_k \ne 0$; $k \ge 1$ since $u_0 \ne 0$. Since $u_k = \sum_{i<k} u_i$, the equation $\sum_{i<k} u_i c_i = -u_k c_k$ forces equality in $|\sum_{i<k} u_i c_i| \le \sum_{i<k} u_i$, i.e. $c_i = -c_k$ for all $i < k$. (b) Using $\sum_{i=1}^{k-1} i 2^{i-1} = (k-2)2^{k-1} + 1$ one finds $\sum_{i<k} e_i u_i = a + \sum_{i=1}^{k-1}(a - 1 - i)2^{i-1} = (a-k+1)2^{k-1}$, and $e_k u_k = (a-k-1)2^{k-1}$. □

4 Powers of two

Proof of Theorem A. For $N = 1$ the sum $\sum_{n \le N} f(n) = f(1)$ does not vanish, so let $N = 2^a$ with $a \ge 1$ and let f be N-periodic, ± 1-valued, with $\sum_{n \le N} f(n) = 0$ and $L(1, f) = 0$. The layers are $d = 2^i$, $0 \le i \le a$, and we write $d_i = f_{2^i}\colon U_{(i)} \to \{\pm 1\}$; this is a digit tree $\mathbf{d}$ of depth a, with $S_{2^i} = S_i(\mathbf{d})$.

A primitive character of conductor $m \mid N$, $m > 1$, has $m = 2^j$ with $2 \le j \le a$, and in (A) all d are of the form 2^i, $i \ge j$, with $d/\varphi(d) = 2$ and no Euler factor. So (A) says that $\sum_{i \ge j} \langle d_i, \xi \rangle = 0$ for every primitive ξ modulo 2^j, and by Lemma 3.3 the value function $X_{\mathbf{d}}$ is constant. By Lemma 3.2(a) all digits are then constant: $d_i \equiv c_i \in \{\pm 1\}$, and $S_{2^i} = u_i c_i$. Condition (C) reads $\sum_i u_i c_i = 0$, so by Lemma 3.5(a) $c_i = s$ for $i < a$ and $c_a = -s$ for some $s = \pm 1$. Condition (B) at $r = 2$ reads $\sum_i e_i u_i c_i = 0$; but by Lemma 3.5(b) with $k = a$ the left-hand side is $s\,2^a \ne 0$. This contradicts Theorem 2.1. □

5 Periods $2^a p$: the master equation and the contraction

Throughout this section p is an odd prime, $a \geq 1$, $N = 2^a p$, and f is N-periodic. The layers are $d = 2^i$ (the residues divisible by p) and $d = 2^i p$ (the residues prime to p), $0 \leq i \leq a$. For $x \in U_{(i)}$ and $y \in U_p$ put

$$g_i(x) = f(2^{a-i} p\, x), \qquad \phi_i(x,y) = f(2^{a-i} z), \quad z \in U_{2^i p},\ z \equiv x\ (2^i),\ z \equiv y\ (p),$$

so that $g_i = f_{2^i}$ and $\phi_i(x,y) = f_{2^i p}(z)$. On $U_{(a)}$ define the value functions

$$\widehat{G}(u) = \sum_{i=0}^{a} 2^i g_i(u \bmod 2^i), \qquad \widehat{\Phi}_y(u) = \sum_{i=0}^{a} 2^i \phi_i(u \bmod 2^i, y), \qquad Z(u) = \frac{1}{p-1} \sum_{y \in U_p} \widehat{\Phi}_y(u).$$

If f is ± 1-valued, (g_i) and, for each y, $(\phi_i(\cdot, y))$ are digit trees of depth a. We write T_r for the operator $(T_r F)(x) = F(rx)$ on functions on a group in which r is a unit.

Proposition 5.1 (master equation)**.** *The conditions* (A) *of Theorem 2.1 for all primitive characters of conductor 2^j or $2^j p$ with $2 \leq j \leq a$ hold if and only if there are $K \in \mathbb{C}$ and $\beta \colon U_p \to \mathbb{C}$ with $\sum_y \beta(y) = 0$ such that for all $u \in U_{(a)}$ and $y \in U_p$*

$$\widehat{\Phi}_y(u) = Z(u) + \beta(y), \tag{2}$$

$$\widehat{G}(u) + p\,\widehat{\Phi}_y(u) - \widehat{\Phi}_y(pu) = K + (p-1)\beta(y). \tag{3}$$

Proof. Conductor 2^j. Let ξ be primitive modulo 2^j. The d in (A) are 2^i and $2^i p$ with $i \geq j$; $d/\varphi(d)$ equals 2 resp. $2p/(p-1)$, and the only Euler factor is $1 - \xi(p)/p$ for $d = 2^i p$. With $h_i(x) = \sum_y \phi_i(x,y)$ we have $\langle f_{2^i p}, \xi \rangle = \sum_{x \in U_{(i)}} h_i(x)\overline{\xi(x)}$ and $(p - \xi(p)) \sum_x h(x)\overline{\xi(x)} = \sum_x (p h(x) - h(px))\overline{\xi(x)}$. After division by 2, (A) for ξ becomes $\sum_{i \geq j} \sum_{x \in U_{(i)}} A_i(x)\overline{\xi(x)} = 0$ with $A_i = g_i + (p\,h_i - T_p h_i)/(p-1)$. By Lemma 3.3 these conditions ($2 \leq j \leq a$) hold iff $\widehat{G} + (p\widehat{H} - T_p \widehat{H})/(p-1)$ is constant, where $\widehat{H} = \sum_y \widehat{\Phi}_y = (p-1)Z$; that is, iff

$$\widehat{G}(u) + pZ(u) - Z(pu) = K \qquad (u \in U_{(a)}) \tag{4}$$

for a constant K.

Conductor $2^j p$. The primitive characters are $\psi\xi$ with $\psi \neq 1$ modulo p and ξ primitive modulo 2^j. The d in (A) are $2^i p$, $i \geq j$, with $d/\varphi(d) = 2p/(p-1)$ and no Euler factor, and $\langle f_{2^i p}, \psi\xi \rangle = \sum_{x \in U_{(i)}} A_i^{\psi}(x)\overline{\xi(x)}$ with $A_i^{\psi}(x) = \sum_y \phi_i(x,y)\overline{\psi(y)}$. By Lemma 3.3 these conditions hold iff $B_\psi(u) = \sum_y \widehat{\Phi}_y(u)\overline{\psi(y)}$ is independent of u for every $\psi \neq 1$. By Fourier inversion on U_p this is equivalent to (2) with $\beta(y) = \frac{1}{p-1} \sum_{\psi \neq 1} B_\psi \psi(y)$.

Finally, given (2), the left-hand side of (3) equals $\widehat{G}(u) + pZ(u) - Z(pu) + (p-1)\beta(y)$, so (3) is equivalent to (4). □

Proposition 5.2 (contraction)**.** *Let f be ± 1-valued and satisfy (3) for some constants K_y in place of $K + (p-1)\beta(y)$, and assume $2^{a-1} \leq p - \chi_4(p)$. Then $\widehat{G}$ and every $\widehat{\Phi}_y$ are constant on $U_{(a)}$; equivalently, all digits $g_i(x)$ and $\phi_i(x,y)$ are independent of $x \in U_{(i)}$.*

Proof. For $a = 1$ there is nothing to prove, so let $a \geq 2$. For $1 \leq j \leq a$ consider the statement

$$Q(j)\colon \quad \widehat{G}(u) \text{ and all } \widehat{\Phi}_y(u) \text{ depend only on } u \bmod 2^j.$$

$Q(a)$ is trivial and $Q(1)$ is the claim, so it suffices to show $Q(j) \Rightarrow Q(j-1)$ for $2 \leq j \leq a$. Assume $Q(j)$ for some $2 \leq j \leq a$, fix y, and define on $U_{(j)}$

$$\delta(u) = \widehat{\Phi}_y(u) - \widehat{\Phi}_y(u + 2^{j-1}), \qquad \gamma(u) = \widehat{G}(u) - \widehat{G}(u + 2^{j-1}).$$

Since $p(u+2^{j-1}) \equiv pu + 2^{j-1} \pmod{2^j}$, subtracting (3) at u and at $u+2^{j-1}$ gives

$$\delta(pu) = p\,\delta(u) + \gamma(u) \qquad (u \in U_{(j)}). \tag{5}$$

The units u and $u+2^{j-1}$ agree modulo 2^{j-1}, so their digits on the levels $i<j$ agree, and each level $i \geq j$ contributes 0 or $\pm 2^{i+1}$. Hence $\delta(u) \in 2^{j+1}\mathbb{Z}$ and $|\gamma(u)| \leq \sum_{i=j}^{a} 2^{i+1} = 2^{a+2} - 2^{j+1}$.

If $j \geq 3$, or $j=2$ and $p \equiv 1 \pmod 4$, Lemma 3.4 (with $\pi = T_p$ on $U_{(j)}$, $q=p$) gives $|\delta| \leq (2^{a+2} - 2^{j+1})/(p-1)$, which is $< 2^{j+1}$ as soon as $2^{a+1-j} < p$. For $j \geq 3$ this follows from $2^{a-2} \leq (p+1)/2 < p$, and for $j=2$, $p \equiv 1 \pmod 4$, it is the hypothesis $2^{a-1} \leq p-1$. If $j=2$ and $p \equiv 3 \pmod 4$, then $U_{(2)} = \{1,3\}$, $p \cdot 1 \equiv 3$ and $\delta(3) = -\delta(1)$, so (5) at $u=1$ reads $-(p+1)\delta(1) = \gamma(1)$ and $|\delta(1)| \leq (2^{a+2}-8)/(p+1) < 8$ by the hypothesis $2^{a-1} \leq p+1$. In all cases $\delta \in 2^{j+1}\mathbb{Z}$ forces $\delta = 0$, and then $\gamma = 0$ by (5). Hence $\widehat{\Phi}_y$ and $\widehat{G}$ are invariant under $u \mapsto u + 2^{j-1}$, which is $Q(j-1)$. The last assertion follows from Lemma 3.2(a). □

Proposition 5.3 (bottom step)**.** *Let f be N-periodic and ± 1-valued, and assume that all digits $g_i(x)$, $\phi_i(x,y)$ are independent of x; write g_i and $\phi_i(y)$. Then the conditions* (C)*,* (B) *at 2 and p, and* (A) *for the characters of conductor p cannot all hold.*

Proof. Suppose they hold. The level sums are $S_{2^i} = u_i g_i$ and $S_{2^i p} = u_i \sum_y \phi_i(y)$. Since $\varepsilon_p(2^i) = 1$ and $\varepsilon_p(2^i p) = -1/(p-1)$, (C) and (B) at p give

$$\sum_i u_i g_i = 0, \qquad \sum_y \Big(\phi_0(y) + Q(y)\Big) = 0, \qquad Q(y) := \sum_{i=1}^{a} u_i \phi_i(y). \tag{6}$$

By Lemma 3.5(a) the first equation gives $g_i = s$ for $i < a$ and $g_a = -s$, with $s = \pm 1$.

For a non-trivial ψ modulo p, the d in (A) are p and $2^i p$ ($i \geq 1$), with weights $p/(p-1)$ and $2p/(p-1)$ and the Euler factor $1 - \psi(2)/2$ for $i \geq 1$; moreover $\langle f_p, \psi\rangle = \sum_y \phi_0(y)\overline{\psi(y)}$ and $\langle f_{2^i p}, \psi\rangle = u_i \sum_y \phi_i(y)\overline{\psi(y)}$. Since $(2-\psi(2)) \sum_y Q(y)\overline{\psi(y)} = \sum_y (2Q(y) - Q(2y))\overline{\psi(y)}$, (A) for all $\psi \neq 1$ says that

$$\phi_0(y) + 2Q(y) - Q(2y) = c \qquad (y \in U_p)$$

for a constant c. Put $X = Q - c$; then $X(2y) = 2X(y) + \phi_0(y)$, and Lemma 3.4 (with $\pi = T_2$ on U_p, $q=2$) gives $|X| \leq 1$. Each $Q(y) = \phi_1(y) + \sum_{i \geq 2} 2^{i-1}\phi_i(y)$ is odd, so $X(2y) - X(y) = Q(2y) - Q(y)$ is even; as it equals $X(y) + \phi_0(y)$, every $X(y)$ is an odd integer, hence $X(y) = \pm 1$. Then $X(2y) = 2X(y) + \phi_0(y) \in \{\pm 1\}$ forces $\phi_0(y) = -X(y)$. Now the second equation in (6) reads $\sum_y (-X(y) + X(y) + c) = (p-1)c = 0$, so $c = 0$ and $Q(y) = X(y) =: \sigma_y \in \{\pm 1\}$. Since $Q(y) = \sum_{k=0}^{a-1} 2^k \phi_{k+1}(y)$, Lemma 3.2(a) with $a-1$ in place of a (for $a = 1$ simply $Q = \phi_1$) gives $\phi_a(y) = \sigma_y$ and $\phi_i(y) = -\sigma_y$ for $1 \leq i < a$; also $\phi_0(y) = -\sigma_y$.

Finally (B) at 2: since $\varepsilon_2(2^i) = \varepsilon_2(2^i p) = e_i$, it reads $\sum_i e_i u_i (g_i + \sum_y \phi_i(y)) = 0$. By Lemma 3.5(b) with $k = a$, $\sum_i e_i u_i g_i = s\,2^a$ and $\sum_i e_i u_i \phi_i(y) = -\sigma_y 2^a$. Hence $\sum_{y \in U_p} \sigma_y = s$. The left side is a sum of the even number $p-1$ of signs, hence even, while s is odd: a contradiction. □

Proof of Theorem B. If $2^a p \in \mathcal{D}$, take f as in Definition 1.1. By Theorem 2.1, f satisfies (A), (B) and (C); by Proposition 5.1 it satisfies (2) and (3); by Proposition 5.2 its digits are independent of x; and Proposition 5.3 gives a contradiction. The reformulation follows because $a < \lfloor \log_2(p - \chi_4(p)) \rfloor + 2$ iff $2^{a-1} \leq p - \chi_4(p)$. □

Remark 5.4. (1) The proof uses (A) only for conductors p, 2^j and $2^j p$, and (B) only at 2 and p. (2) The two contractions have different sources: the first comes from the Euler factor $1 - \xi(p)/p$ of the odd prime in the conductor-2^j conditions and acts on 2-adic differences; the second from the Euler factor $1 - \psi(2)/2$ of the prime 2 in the conductor-p conditions and acts on the residues modulo p. The final parity count is the 2-adic shadow of the logarithmic condition at 2. (3) For $p \equiv 3 \pmod 4$ the step $j=2$ gains a factor $(p+1)/(p-1)$ because T_p swaps the two classes modulo 4; this is the source of $\chi_4(p)$ in the statement, and Theorem C shows that the bound is attained for $p = 3$. (4) For $a = 1$ the contraction is vacuous and the proof is the case $N = 2p$ of the theorem of Murty and Pathak [12].

6 Digit-tree constructions and exact thresholds

We now run the digit trees backwards. For odd x with $|x| < 2^{a+1}$ let $\lambda_a(x) = \sum_{i=0}^{a} w_i\, d_i(x)$, where

$$w_0 = a\,2^{1-a}, \qquad w_i = (a-i-1)\,2^{i-a} \quad (1 \le i \le a),$$

so that $w_i = e_i u_i 2^{1-a}$.

Lemma 6.1. *For a digit tree* $\mathbf{d}$ *of depth* a *with value function* X*:*

(a) $\sum_{i=0}^{a} S_i(\mathbf{d}) = \frac{1}{2}(\overline{X} - \chi_4(X))$*, where* $\overline{X} = 2^{1-a}\sum_{u\in U_{(a)}} X(u)$ *is the mean of* X *and* $\chi_4(X)$ *is the common value of* $\chi_4(X(u))$*;*

(b) $\sum_{i=0}^{a} e_i S_i(\mathbf{d}) = \sum_{u\in U_{(a)}} \lambda_a(X(u))$.

Proof. A point of $U_{(i)}$ has $2^{a-1}/u_i$ preimages in $U_{(a)}$, so $S_i = u_i 2^{1-a}\sum_u d_i(u \bmod 2^i)$. This gives (b) at once, and (a) because $\sum_i u_i d_i = \frac{1}{2}(X + d_0)$ and $d_0 = -\chi_4(X)$ (Lemma 3.2(c)). □

Lemma 6.2 (pairing lemma)**.** *Let* $\varepsilon \in \{\pm 1\}$ *and* $t \in \mathbb{Z}$*,* $t \neq 0$*, with* $4|t| + 1 < 2^{a+1}$*. Then*

$$\lambda_a(\varepsilon + 4t) + \lambda_a(\varepsilon - 4t) = -\varepsilon\, 2^{\,v_2(t)+3-a}, \qquad \text{and} \qquad \lambda_a(\varepsilon) = -2\varepsilon.$$

Proof. For odd x with $|x| < 2^{a+1}$ let $x_k = \sum_{i<k} 2^i d_i(x)$; then x_k is odd, $|x_k| < 2^k$, and $x \equiv x_k + 2^k \pmod{2^{k+1}}$, so x_k is the unique odd number in $(-2^k, 2^k)$ with this property. Since $\sum_{k=1}^{a} x_k = \sum_{i<a}(a-i)2^i d_i$, a direct computation gives

$$2^a \lambda_a(x) = \sum_{k=1}^{a} x_k - x + (a+1)d_0(x). \tag{7}$$

Let $x = \varepsilon + 4t$, $y = \varepsilon - 4t$ and $v = v_2(t) \le a-2$. Then $d_0(x) = d_0(y) = -\varepsilon$ and $x_1 = y_1 = -\varepsilon$. For $k \ge 2$, $x_k + y_k \equiv x + y \equiv 2\varepsilon \pmod{2^{k+1}}$ and $|x_k + y_k| < 2^{k+1}$, so $x_k + y_k \in \{2\varepsilon,\ 2\varepsilon - \varepsilon 2^{k+1}\}$, and the second value occurs iff $x_k = y_k = \varepsilon(1-2^k)$, i.e. iff $x \equiv \varepsilon \pmod{2^{k+1}}$, i.e. iff $k \le v+1$. Summing, $\sum_k (x_k + y_k) = -2\varepsilon + 2\varepsilon(a-1) - \varepsilon(2^{v+3} - 8)$, and (7) gives $2^a(\lambda_a(x) + \lambda_a(y)) = -\varepsilon 2^{v+3}$. For $t = 0$ the same computation with $x = y = \varepsilon$ (every $k \ge 2$ of the second kind) gives $2^{a+1}\lambda_a(\varepsilon) = -\varepsilon 2^{a+2}$. □

Theorem E (digit-tree criterion)**.** *Let* p *be an odd prime,* $a \ge 2$*,* $\varepsilon \in \{\pm 1\}$*, and let* $H' \colon U_{(a)} \to \mathbb{Z}$ *satisfy*

(i) $H'(-u) = -H'(u)$*, and* $H'(u) \equiv H'(u') \pmod{2^k}$ *whenever* $u \equiv u' \pmod{2^k}$*,* $1 \le k \le a$*;*

(ii) $H'(u) \neq 0$ *and* $G'(u) := H'(pu) - pH'(u) \neq 0$ *for all* u*;*

(iii) $4\max|H'| + 1 < 2^{a+1}$ *and* $4\max|G'| + 1 < 2^{a+1}$*;*

(iv) $(\star)$ $\displaystyle\sum_{u\in U_{(a)}} 2^{v_2(G'(u))} = (p-1)\sum_{u\in U_{(a)}} 2^{v_2(H'(u))}$.

Let $H = \varepsilon + 4H'$*,* $G = -\varepsilon + 4G'$*, and let* (η_i)*,* (g_i) *be the digit trees with value functions* H*,* G *(Lemma 3.2(c)). For* $n \in \mathbb{Z}$ *let* $i = a - \min(v_2(n), a)$ *and let* $x \in U_{(i)}$ *be the residue modulo* 2^i *of the odd part of* n *if* $p \nmid n$*, and of the odd part of* n/p *if* $p \mid n$*. Define the* $2^a p$*-periodic function*

$$f(n) = \eta_i(x) \quad \text{if } p \nmid n, \qquad f(n) = g_i(x) \quad \text{if } p \mid n.$$

Then $\sum_{n\le 2^a p} f(n) = 0$ *and* $\sum_{n\ge 1} f(n)/n = 0$*; in particular* $2^a p \in \mathcal{D}$*.*

Proof. By (i) and (iii), H and G satisfy the conditions of Lemma 3.2(c): they are odd, bounded by $2^{a+1}-1$, and $H(u)-H(u')=4(H'(u)-H'(u'))\equiv 0 \pmod{2^{k+2}}$ when $u\equiv u' \pmod{2^k}$; the same holds for G' because multiplication by p preserves congruences modulo 2^k. In the notation of Section 5, $\phi_i(x,y)=\eta_i(x)$ for all y, so $\widehat{\Phi}_y=H$, $Z=H$, $\beta=0$, and

$$\widehat{G}+pZ-Z\circ T_p=-\varepsilon+4G'+p(\varepsilon+4H')-(\varepsilon+4H'\circ T_p)=(p-2)\varepsilon$$

is constant; by Proposition 5.1, (A) holds for all conductors 2^j and 2^jp. For conductor p, $\langle f_{2^ip},\psi\rangle=\sum_x\eta_i(x)\cdot\sum_y\overline{\psi(y)}=0$ and the layers 2^i do not see ψ; so (A) holds for all characters. Since H' is odd, $\overline{H'}=0=\overline{G'}$, hence $\overline{H}=\varepsilon=\chi_4(H)$ and $\overline{G}=-\varepsilon=\chi_4(G)$, and Lemma 6.1(a) shows that the level sums of both trees add up to 0. As $S_{2^i}=S_i(g)$ and $S_{2^ip}=(p-1)S_i(\eta)$, this gives (C) and (B) at p (whose coefficients are 1 and $-1/(p-1)$). Finally (B) at 2 reads $\sum_i e_i(S_i(g)+(p-1)S_i(\eta))=0$, i.e. by Lemma 6.1(b)

$$\sum_u\lambda_a(G(u))+(p-1)\sum_u\lambda_a(H(u))=0.$$

Pair u with $-u$: $H(\pm u)=\varepsilon\pm 4H'(u)$ and $G(\pm u)=-\varepsilon\pm 4G'(u)$. By Lemma 6.2 the left-hand side equals $\varepsilon 2^{2-a}\bigl(\sum_u 2^{v_2(G'(u))}-(p-1)\sum_u 2^{v_2(H'(u))}\bigr)$, which vanishes by $(\star)$. By Theorem 2.1 (sufficiency), $L(1,f)=0$. □

Remark 6.3. If H' depends only on u mod 2^J, both sides of $(\star)$ are 2^{a-J} times the corresponding sums over $U_{(J)}$, so $(\star)$ can be checked on $U_{(J)}$, and a is then only constrained by (iii). Theorem E is the exact converse of the reduction of Section 5 for functions that depend on n mod p only through the condition $p\mid n$: for such f, Proposition 5.1, Lemma 6.1 and Lemma 6.2 show that, when H' is odd, H' and G' do not vanish, and $\varepsilon_G=-\varepsilon_H$, condition $(\star)$ is equivalent to (B) at 2.

Proof of Theorem C. Fermat primes. Let $p=2^v+1$ with $v=2^k\geq 2$, so $p\equiv 1 \pmod 4$. Take $H'=-\chi_4$ (on $U_{(a)}$ via u mod 4). It is odd and 1-Lipschitz, and $G'(u)=-\chi_4(pu)+p\chi_4(u)=(p-1)\chi_4(u)=2^v\chi_4(u)$, so $(\star)$ reads $2^{a-1}2^v=(p-1)2^{a-1}$, which holds. Condition (iii) asks $4\cdot 2^v+1<2^{a+1}$, i.e. $a\geq v+2$. By Theorem E, $2^{v+2}p\in\mathcal{D}$, while Theorem B gives $2^{v+1}p\notin\mathcal{D}$ since $2^v\leq p-1$. Hence $a_0(p)=v+2=2^k+2$.

$p=3$. Take $a=4$, $\varepsilon=1$ and H' on $U_{(3)}$ given by $H'(1)=H'(3)=-2$, $H'(5)=H'(7)=2$. It is odd and 1-Lipschitz; $G'(u)=H'(3u)-3H'(u)$ takes the values $4,4,-4,-4$ at $u=1,3,5,7$. On $U_{(3)}$ the left side of $(\star)$ is $4\cdot 2^2=16$ and the right side is $(3-1)\cdot 4\cdot 2^1=16$, and (iii) holds for $a=4$ since $4\cdot 4+1<32$. Thus $48\in\mathcal{D}$, and $24\notin\mathcal{D}$ by Theorem B ($2^2\leq 3+1$). Hence $a_0(3)=4$. □

Example 6.4 (the sawtooth for $p=13$)**.** Let $H'(u)=r_3(u)-4$ where $r_3(u)\in\{1,3,5,7\}$ is u mod 8, i.e. $H'=(-3,-1,1,3)$ on $(1,3,5,7)$; with the opposite sign this is the certificate in Table 2. For $p=13\equiv 5 \pmod 8$, $G'=(40,16,-16,-40)$, so $(\star)$ reads $8+16+16+8=12\cdot 4$, and (iii) holds for $a=7$. Hence $2^7\cdot 13=1664\in\mathcal{D}$. More generally $H'(u)=r_J(u)-2^{J-1}$ satisfies $(\star)$ iff $\sum_r 2^{v_2(m-\lfloor pr/2^J\rfloor)}=m$ with $m=(p-1)/2$, the sum running over the odd $r<2^J$.

Table 2 lists, for the primes $p<60$, the lower bound of Theorem B and the smallest depth produced by Theorem E with a function H' found by a small search (Section 9); each certificate was verified with exact integer arithmetic (digit trees, level sums and condition (B) at 2), and the resulting f for $p\in\{3,5,7,13,17\}$ were also checked against the full system of Theorem 2.1 or numerically (Section 9).

7 Periods $12p$

Let $p\geq 5$ be prime and $N=12p$. For $d\mid 12$ let $c_d(T)=\prod_{r^e\|d} r^{e-1}(r-T_r)$, an operator on functions on U_p; thus $c_1=1$, $c_2=2-T_2$, $c_4=2(2-T_2)$, $c_3=3-T_3$, $c_6=(2-T_2)(3-T_3)$, $c_{12}=2(2-T_2)(3-T_3)$.

Lemma 7.1 (reduction)**.** *Let $s_d \in \{\pm 1\}$ and $\Psi_d \colon U_p \to \{\pm 1\}$ for $d \mid 12$, and define the N-periodic f by*

$$f(n) = \begin{cases} s_d & \text{if } p \mid n, \\ \Psi_d(n/\gcd(n,12) \bmod p) & \text{if } p \nmid n, \end{cases} \qquad d = 12/\gcd(n,12).$$

Put $\sigma_d = \sum_y \Psi_d(y)$. Then $\sum_{n \le N} f(n) = 0$ and $L(1,f) = 0$ if and only if

(R1) $\sum_{d \mid 12} c_d(T)\Psi_d = 0$ *on* U_p;

(R2) $\sum_d \varphi(d) s_d = 0$ *and* $\sum_d \varphi(d)\sigma_d = 0$;

(R3) $\sum_d \varphi(d)\,\epsilon_r(d)(s_d + \sigma_d) = 0$ *for* $r = 2, 3$, *where* $\epsilon_r(d) = v_r(12/d) - [\,r \mid d\,]/(r-1)$.

Proof. The layers of N are $D = d$ and $D = dp$ with $d \mid 12$; on the layer d (the residues divisible by p) $f = s_d$, and on the layer dp, $f_{dp}(x) = \Psi_d(x \bmod p)$ for $x \in U_{dp} \cong U_d \times U_p$. If a primitive character χ has conductor with non-trivial 12-part, its 12-part is non-principal modulo d on every layer it meets, and all $\langle f_D, \chi\rangle$ vanish because f_D does not depend on the U_d-coordinate. For $\chi = \psi$ of conductor p, the term of the layer dp in (A) is

$$\frac{p}{p-1} \cdot \frac{d}{\varphi(d)} \prod_{r \mid d} \Big(1 - \frac{\psi(r)}{r}\Big) \cdot \varphi(d) \langle \Psi_d, \psi \rangle = \frac{p}{p-1} \langle c_d(T)\Psi_d, \psi \rangle,$$

so (A) is equivalent to $\sum_d c_d(T)\Psi_d$ being constant; summing over U_p, the constant is $\frac{1}{p-1}\sum_d \varphi(d)\sigma_d$. Next $S_d = \varphi(d)s_d$, $S_{dp} = \varphi(d)\sigma_d$, $\varepsilon_p(d) = 1$, $\varepsilon_p(dp) = -1/(p-1)$ and $\varepsilon_r(d) = \varepsilon_r(dp) = \epsilon_r(d)$ for $r \in \{2,3\}$; so (C) and (B) at p are equivalent to (R2), (B) at $2, 3$ to (R3), and under (R2) the constant above is 0. □

Let $\lambda = (\frac{\cdot}{p})$ and $\chi(2) = (\frac{2}{p})$, $\chi(3) = (\frac{3}{p})$, so that $T_r\lambda = \chi(r)\lambda$. A *certificate* consists of signs ϵ_d, s_d ($d \mid 12$), a sign η, and finite sets $B_d \subset \mathbb{Z}^2_{\ge 0}$, and we put, for a fixed $y_0 \in U_p$ with $\lambda(y_0) = \eta$,

$$\Psi_d(y) = \epsilon_d\,\lambda(y)\big(1 - 2 \cdot [\,y \in P_d\,]\big), \qquad P_d = \{2^i 3^j y_0 : (i,j) \in B_d\} \subset U_p. \tag{8}$$

Proposition 7.2. *Suppose that the points $2^i 3^j y_0$, $(i,j) \in B_d$, are pairwise distinct modulo p for each d, and that*

(L0) $\sum_d \epsilon_d\, c_d(\chi) = 0$, *where $c_d(\chi)$ is $c_d(T)$ with T_r replaced by $\chi(r)$;*

(L1) $\sum_d \epsilon_d\, c_d(\chi(2)X^{-1}, \chi(3)Y^{-1})\, P_d(X,Y) = 0$ *in the Laurent polynomial ring $\mathbb{Z}[X^{\pm 1}, Y^{\pm 1}]$, where $P_d(X,Y) = \sum_{(i,j)\in B_d} X^i Y^j$ and $c_d(\cdot,\cdot)$ denotes $c_d(T)$ with T_2, T_3 replaced by the two arguments;*

(L2) (R2) *and* (R3) *hold with* $\sigma_d = -2\epsilon_d\eta\, P_d(\chi(2), \chi(3))$.

Then the function f of Lemma 7.1 built from (8) satisfies $\sum_{n \le 12p} f(n) = 0$ and $L(1,f) = 0$.

Proof. By distinctness, $1_{P_d}(y) = \#\{(i,j) \in B_d : 2^i 3^j y_0 = y\}$, and for $m = 2^\alpha 3^\beta$ we have $T_m(\lambda 1_{P_d}) = \chi(m)\lambda\, 1_{m^{-1}P_d}$, where $m^{-1}P_d$ corresponds to the monomials of $X^{-\alpha}Y^{-\beta}P_d$. Hence the image of (L1) under $X^iY^j \mapsto [2^i 3^j y_0] \in \mathbb{Z}[U_p]$ says $\sum_d \epsilon_d c_d(T)(\lambda 1_{P_d}) = 0$, and together with (L0), $\sum_d c_d(T)\Psi_d = \lambda \sum_d \epsilon_d c_d(\chi) - 2\sum_d \epsilon_d c_d(T)(\lambda 1_{P_d}) = 0$, which is (R1). Since $\sum_y \lambda(y) = 0$, $\sigma_d = -2\epsilon_d \sum_{y \in P_d} \lambda(y) = -2\epsilon_d \eta P_d(\chi(2), \chi(3))$, so (L2) gives (R2) and (R3). Lemma 7.1 concludes. □

$(\chi(2),\chi(3))$	$(-1,-1)$	$(-1,+1)$	$(+1,-1)$
$p \bmod 24$	$5, 19$	$11, 13$	$7, 17$
$(\epsilon_1,\epsilon_2,\epsilon_3,\epsilon_4,\epsilon_6,\epsilon_{12})$	$(+,-,-,-,-,+)$	$(-,+,-,-,-,+)$	$(-,-,+,+,+,-)$
$(s_1,s_2,s_3,s_4,s_6,s_{12})$	$(+,+,+,-,+,-)$	$(-,-,-,-,+,+)$	$(+,+,+,-,+,-)$
η	-1	$+1$	-1
P_1	$Y^2+XY^2+X^2Y+X^2Y^2$	$Y+X+XY+X^2+X^2Y$	$Y+X+XY$
P_2	XY^2+X^2Y	$XY+X^2+X^2Y$	X
P_3	X^2Y^2	$XY+X^2Y$	XY
P_4	X^2Y^2	X^2Y	$X+XY$
P_6	X^2Y^2	X^2Y	XY
P_{12}	X^2Y^2	X^2Y	XY
$(\sigma_1,\sigma_2,\sigma_3,\sigma_4,\sigma_6,\sigma_{12})$	$(0,4,-2,-2,-2,2)$	$(2,-2,0,2,2,-2)$	$(2,-2,-2,0,-2,2)$
distinctness fails only for	$p\in\{2,3\}$	$p\in\{2,3,5\}$	$p\in\{2,3\}$

Table 1: The three certificates for $N=12p$ (Proposition 7.2). Signs $\pm$ stand for ±1.

Proof of Theorem D. Let $p\ge5$, $p\not\equiv\pm1 \pmod{24}$. Then $(\chi(2),\chi(3))\neq(1,1)$, and p lies in one of the three columns of Table 1 (the class $(-1,1)$ contains no prime below 11). We check (L0)–(L2) column by column. Write $A=2-\chi(2)X^{-1}$ and $B=3-\chi(3)Y^{-1}$, so that in (L1) the operators $c_1,c_2,c_4,c_3,c_6,c_{12}$ become $1,A,2A,B,AB,2AB$. Vectors indexed by d are listed in the order $d=1,2,3,4,6,12$; we use $(\varphi(d))_d=(1,1,2,2,2,4)$, $(\varphi(d)\epsilon_2(d))_d=(2,0,4,-2,0,-4)$ and $(2\varphi(d)\epsilon_3(d))_d=(2,2,-2,4,-2,-4)$.

Column $(1,-1)$: $A=2-X^{-1}$, $B=3+Y^{-1}$. (L0): $c_d(\chi)=(1,1,4,2,4,8)$ and $-1-1+4+2+4-8=0$. (L1) reads $-(Y+X+XY)-AX+BXY+2A(X+XY)-ABXY=0$; expanding, $-AX=-2X+1$, $BXY=3XY+X$, $2A(X+XY)=4X+4XY-2-2Y$ and $-ABXY=-6XY-2X+3Y+1$, and the coefficients of $XY,X,Y,1$ add up to $-1+3+4-6$, $-1-2+1+4-2$, $-1-2+3$, $1-2+1$, all 0. With $\eta=-1$, $\sigma=(2,-2,-2,0,-2,2)$ and $s+\sigma=(3,-1,-1,-1,-1,1)$; (R2): $1+1+2-2+2-4=0$ and $2-2-4+0-4+8=0$; (R3): $6+0-4+2+0-4=0$ and $6-2+2-4+2-4=0$. The points of $P_1=\{3,2,6\}y_0$ and $P_4=\{2,6\}y_0$ are distinct for $p\ge5$.

Column $(-1,-1)$: $A=2+X^{-1}$, $B=3+Y^{-1}$. (L0): $c_d(\chi)=(1,3,4,6,12,24)$ and $1-3-4-6-12+24=0$. Since $P_3=P_4=P_6=P_{12}=X^2Y^2$, (L1) reads $P_1-AP_2+(AB-2A-B)X^2Y^2=0$; here $(AB-2A-B)X^2Y^2=XY+XY^2+X^2Y-X^2Y^2$ and $AP_2=2XY^2+2X^2Y+Y^2+XY$, so the left side is $(Y^2+XY^2+X^2Y+X^2Y^2)-(2XY^2+2X^2Y+Y^2+XY)+(XY+XY^2+X^2Y-X^2Y^2)=0$. With $\eta=-1$, $\sigma=(0,4,-2,-2,-2,2)$ and $s+\sigma=(1,5,-1,-3,-1,1)$; (R2): $1+1+2-2+2-4=0$ and $0+4-4-4-4+8=0$; (R3): $2+0-4+6+0-4=0$ and $2+10+2-12+2-4=0$. The points of $P_1=\{9,18,12,36\}y_0$ and $P_2=\{18,12\}y_0$ are distinct for $p\ge5$.

Column $(-1,1)$: $A=2+X^{-1}$, $B=3-Y^{-1}$. (L0): $c_d(\chi)=(1,3,2,6,6,12)$ and $-1+3-2-6-6+12=0$. Since $P_4=P_6=P_{12}=X^2Y$, (L1) reads $-P_1+AP_2-BP_3+A(B-2)X^2Y=0$; here $AP_2=3XY+2X^2+2X^2Y+X+Y$, $BP_3=3XY+3X^2Y-X-X^2$ and $A(B-2)X^2Y=2X^2Y+XY-2X^2-X$, and the coefficients of XY,X^2,X^2Y,X,Y on the left are $-1+3-3+1$, $-1+2+1-2$, $-1+2-3+2$, $-1+1+1-1$, $-1+1$, all 0. With $\eta=1$, $\sigma=(2,-2,0,2,2,-2)$ and $s+\sigma=(1,-3,-1,1,3,-1)$; (R2): $-1-1-2-2+2+4=0$ and $2-2+0+4+4-8=0$; (R3): $2+0-4-2+0+4=0$ and $2-6+2+4-6+4=0$. The points of $P_1=\{3,2,6,4,12\}y_0$, $P_2=\{6,4,12\}y_0$ and $P_3=\{6,12\}y_0$ are distinct for $p\ge7$, and this class contains no prime below 11.

All these identities were also verified symbolically (Section 9). Proposition 7.2 gives $12p\in\mathcal{D}$.

Minimality: a proper divisor M of $12p$ with $M\in\mathcal{D}$ would satisfy $4\mid M$ (Lemma 2.2), so $M\in\{4,12,4p\}$, and these are excluded by Theorem A and by Theorem B with $a=2$ (note $2\le p-\chi_4(p)$ for all odd p). □

Remark 7.3. For $(\chi(2),\chi(3))=(1,1)$ the Legendre symbol is invariant under T_2 and T_3; no

certificate of the shape (8) with small boxes exists, and certificates with a constant base on some layer are impossible (Section 9). Nevertheless $12p \in \mathcal{D}$ for all 10 primes $p < 250$ in this class (Section 9); see Conjecture 10.2.

8 Consequences

Proof of Corollary 1.4. (ii) By Lemma 2.2(b) the elements of $\mathcal{D}$ are multiples of 4. Among $4, 8, \dots, 32$, the powers of 2 are excluded by Theorem A, and $12 = 4 \cdot 3$, $20 = 4 \cdot 5$, $28 = 4 \cdot 7$ ($a = 2$) and $24 = 8 \cdot 3$ ($a = 3$, $2^2 \le 3 + 1$) by Theorem B. The function of Appendix A shows $36 \in \mathcal{D}$ (Tengely's period).

(i) Among the multiples of 4 below 48 the non-squares other than those in (ii) are $40 = 8 \cdot 5$ ($2^2 \le 5 - 1$) and $44 = 4 \cdot 11$, excluded by Theorem B. $48 \in \mathcal{D}$ by Theorem C. The numbers $2^a \cdot 3$ ($a \ge 4$) and $2^a p$ with $p \ge 5$ a Fermat prime and $2^{a-1} > p - 1$ have $v_p = 1$, and $v_p(12p) = 1$ for $p \ge 5$, so they are not squares; they lie in $\mathcal{D}$ by Theorems C and D and Lemma 2.2(a).

(iii) By Theorem D every $12p$ with $p \ge 5$, $p \not\equiv \pm 1 \pmod{24}$, is a minimal element of $\mathcal{D}$, and there are infinitely many such primes (Dirichlet). If $\mathcal{D}$ were the set of multiples of finitely many integers, it would have only finitely many minimal elements.

(iv) This is Lemma 2.2(c) combined with Theorems A and B. □

Corollary 8.1. *If $2^a p \in \mathcal{D}$ then $p - \chi_4(p) < 2^{a-1}$; in particular $p < 2^{a-1}$, so for each a only finitely many primes occur. For every odd prime p, $a_0(p) \ge \log_2 p + 1$; for $p = 3$ and for Fermat primes p, $a_0(p) = \lfloor \log_2(p - \chi_4(p)) \rfloor + 2$.*

Proof. The first statement is Theorem B. It gives $p - 1 \le p - \chi_4(p) < 2^{a-1}$, so $a \ge 2$ and $p \le 2^{a-1}$, and $p \ne 2^{a-1}$ because p is odd and 2^{a-1} is even. For the second, $\lfloor \log_2(p - \chi_4(p)) \rfloor + 2 \ge \lfloor \log_2(p-1) \rfloor + 2 \ge \log_2 p + 1$, because $2^{\lfloor \log_2(p-1) \rfloor} > (p-1)/2$ and hence $2^{\lfloor \log_2(p-1) \rfloor} \ge p/2$. The rest is Theorem C. □

9 Computations

None of the computations below is used in the proofs of Theorems A–E. They were made with exact integer and rational arithmetic in Python (SymPy, python-flint), the CP-SAT solver of OR-Tools for searches, and mpmath for numerical values; scripts and logs are in `xxiv/`. Okada's full system was implemented once from Theorem 2.1, with one row per Galois orbit of characters and coordinates in a power basis of the cyclotomic field; its rank was checked to be $\varphi(N) + \omega(N)$ for 18 values of N between 12 and 224.

9.1 Verification of the constructions

- *Theorem E/C.* The functions f produced by Theorem E (implemented independently from its statement) for $N = 48, 80, 896$ satisfy every equation of the full Okada system exactly; for $N = 48, 80, 896, 1088 = 2^6 \cdot 17, 1664 = 2^7 \cdot 13$ the value $|L(1, f)|$ computed from $L(1, f) = -\frac{1}{N} \sum_{n=1}^{N} f(n)\psi(n/N)$ with 40 digits is below 10^{-40}.

- *Theorem D.* For each of the three classes the identities (L0), (L1), (R2), (R3) were verified symbolically (SymPy), together with the list of primes for which distinctness fails. The resulting f satisfy the full Okada system exactly for the 12 primes $5 \le p \le 53$ with $p \not\equiv \pm 1 \pmod{24}$ (19 to 211 equations each), and $|L(1, f)| < 2 \cdot 10^{-31}$ for all 60 such primes $p < 400$.

- *Class* $(1, 1)$. For each of the 10 primes $p < 250$ with $p \equiv \pm 1 \pmod{24}$, namely 23, 47, 71, 73, 97, 167, 191, 193, 239, 241, a CP-SAT search in the reduced system of Lemma 7.1 produced a solution; those with $p \le 97$ satisfy the full Okada system exactly, and all have $|L(1, f)| < 3 \cdot 10^{-31}$. For $p = 263$ the search was inconclusive after 900 s. In this class

no certificate of the shape (8) exists with one orbit and $B_d \subset [0,3]^2$, or with up to three orbits and $B_d \subset [0,2]^2$ (CP-SAT proved infeasibility). Moreover, a p-independent variant in which λ is replaced by the constant 1 on at least one layer cannot exist: valid for infinitely many p, it would force the coefficient of $p-1$ in (R2), (R3) to vanish, i.e. a signed sum $\sum_{d\in\Gamma} \pm(\varphi(d), \varphi(d)\epsilon_2(d), 2\varphi(d)\epsilon_3(d)) = 0$ over the set $\Gamma \neq \emptyset$ of constant layers, together with $\sum_{d\notin\Gamma} \pm\varphi(d) = 0$; the only solution of the first condition is $\Gamma = \{1,2,3,4,6\}$, which violates the second.

- *Lemmas.* Lemma 6.2 was checked for all $2 \le a \le 11$, $\varepsilon = \pm 1$ and all admissible t (8144 cases); Lemma 3.5(b) for $1 \le k \le a \le 15$; and Lemma 3.5(a) exhaustively for $a \le 8$.

- *Proposition 5.1 and 5.2.* For the solutions with $N = 48$ and $N = 80$ found by an unrestricted search, (2) and (3) hold with $K = -(f(0) + \sum_y f(2^a y))$, and the invariance $Q(j) \Rightarrow Q(j-1)$ holds exactly at the levels predicted by the proof ($j = 4$ for $N = 48$; $j = 4, 3$ for $N = 80$) and fails at the next level, as it must since $48, 80 \in \mathcal{D}$.

9.2 The set $\mathcal{D}$ below 333

For every $N \equiv 0 \pmod 4$, $N \le 332$, that is not a multiple of a smaller element found, CP-SAT decided the existence of a ± 1-valued f satisfying the full Okada system (a complete search; infeasibility was proved by the solver). The minimal elements of $\mathcal{D}$ below 333 are

$$36,\ 48,\ 60,\ 80,\ 84,\ 132,\ 156,\ 200,\ 204,\ 220,\ 228,\ 260,\ 276,\ 280,$$

and all other multiples of 4 below 333 that are not multiples of these are outside $\mathcal{D}$. Among the latter, all except $100, 140, 196, 224, 308$ are excluded by Theorems A and B; the five exceptions rely on the computation. The members $60, 84, 132, 156, 204, 228, 276$ are the periods $12p$ for $5 \le p \le 23$. Further complete searches: $340, 364, 460, 476 \notin \mathcal{D}$ and $380, 572 \in \mathcal{D}$ (periods $4pq$; the pattern is irregular), $352 = 2^5 \cdot 11$, $704 = 2^6 \cdot 11$, $416 = 2^5 \cdot 13$, $832 = 2^6 \cdot 13$ and $448 = 2^6 \cdot 7$ are not in $\mathcal{D}$; $532 = 4 \cdot 7 \cdot 19$ was undecided after 2400 s. As a check of Theorem B at depth $a = 5$, complete searches confirm $544 = 2^5 \cdot 17 \notin \mathcal{D}$ and $608 = 2^5 \cdot 19 \notin \mathcal{D}$. The function for $1408 = 2^7 \cdot 11$ in Table 2 satisfies all 642 equations of the full Okada system exactly.

9.3 Thresholds

Table 2 collects what is known about $a_0(p)$ for $p < 60$.

10 Open problems

Conjecture 10.1 (dichotomy)**.** *$a_0(p) < \infty$ for every odd prime p; that is, every odd prime p divides some element $2^a p$ of $\mathcal{D}$.*

Theorem E reduces Conjecture 10.1 to the existence of one odd 1-Lipschitz function H' satisfying ($\star$); such functions were found for every prime $p < 127$: for $p < 60$ see Table 2; every prime $p \equiv 1 \pmod 4$ below 130 already has one among the combinations $c_1\varphi_{J_1} + c_2\varphi_{J_2}$ of two sawtooth or character components ($\varphi_J(u) = r_J(u) - 2^{J-1}$, χ_4, χ_{-8}; `logs/combo_3_130.log`), and the primes $67, 71, 79, 83, 103, 107$ were treated by CP-SAT (depths $12, 12, 13, 12, 12, 13$; `logs/star_list.log`). For the Mersenne prime $p = 127$, where $p - \chi_4(p) = 2^7$ and Theorem B gives $a_0(127) \ge 9$, our searches found no certificate. By Theorem B, $a_0(p) - \log_2 p$ is bounded below; we do not know whether it is bounded above.

Conjecture 10.2. *$12p \in \mathcal{D}$ for every prime $p \ge 5$.*

By Theorem D only the primes $p \equiv \pm 1 \pmod{24}$ remain, and these were verified for $p < 250$.

p	$p-\chi_4(p)$	Thm. B	Thm. E	known $a_0(p)$	certificate H' (values at $u=1,3,\dots,2^J-1$)
3	4	4	4	**4**	$J=3$: $(-2,-2,2,2)$
5	4	4	4	**4**	$J=2$: $(-1,1)$
7	8	5	7	**7**[a]	$J=4$: $(6,-2,2,2,-2,-2,2,-6)$
11	12	5	8	**7**[b]	$J=5$ (see log)
13	12	5	7	**7**[c]	$J=3$: $(3,1,-1,-3)$
17	16	6	6	**6**	$J=2$: $(-1,1)$
19	20	6	9	6–9	$J=5$ (see log)
23	24	6	9	6–9	$J=4$: $(-6,-6,-10,-2,2,10,6,6)$
29	28	6	10	6–10	$J=4$: $(1,3,-11,-1,1,11,-3,-1)$
31	32	7	11	7–11	$J=6$ (see log)
37	36	7	10	7–10	$J=3$: $(-5,-7,7,5)$
41	40	7	9	7–9	$J=4$: $(3,-3,-5,-3,3,5,3,-3)$
43	44	7	10	7–10	$J=5$ (see log)
47	48	7	12	7–12	$J=5$ (see log)
53	52	7	10	7–10	$J=5$ (see log)
59	60	7	11	7–11	$J=6$ (see log)

Table 2: Bounds for $a_0(p)$, $p<60$. Column "Thm. B" is $\lfloor \log_2(p-\chi_4(p))\rfloor+2$; column "Thm. E" is the depth given by the certificate. [a] $224,448\notin\mathcal{D}$ by complete search. [b] $352,704\notin\mathcal{D}$ by complete search, and $1408=2^7\cdot 11\in\mathcal{D}$ by a digit-tree solution of the reduced 2-adic model (a function of n mod 2^7 and of $[11\mid n]$), checked against all 642 equations of the full Okada system. [c] $416,832\notin\mathcal{D}$ by complete search. All certificates are listed in `logs/star_3_60.log`.

Problem 10.3. Determine $\mathcal{D}$. In particular: is $4pq\in\mathcal{D}$ decided by a congruence condition on the odd primes $p<q$? (We found $220,260,380,572\in\mathcal{D}$ and $140,308,340,364,460,476\notin\mathcal{D}$.) Does $\mathcal{D}$ have minimal elements of arbitrarily large 2-adic valuation? By Theorem A and the definition of $a_0(p)$, $2^{a_0(p)}p$ is a minimal element of $\mathcal{D}$ whenever $a_0(p)<\infty$, and $a_0(p)\geq\log_2 p+1$ by Corollary 8.1; so a positive answer follows from Conjecture 10.1 for infinitely many p (for instance $896=2^7\cdot 7$ and $1408=2^7\cdot 11$ are minimal).

A Explicit functions

We list $f(1),f(2),\dots,f(N)$ (so the last entry is $f(0)$), with $+$ for 1 and $-$ for -1. Both functions satisfy the full Okada system exactly, and $|L(1,f)|<10^{-41}$.

$N=36$: `+--+++-+-+-+-+++-+-+--++++-+`

$N=48$ (Theorem E with $p=3$, $a=4$, $\varepsilon=1$, $H'=(-2,-2,2,2)$ on $U_{(3)}$):

`-++++-+-+-++---+-+++-+++-+-++-+++---+-++`

In the second example the value functions are $H=(-7,-7,9,9)$ and $G=(15,15,-17,-17)$ at $u\equiv 1,3,5,7 \pmod 8$, with signed binary digits (levels $0,\dots,4$) $-7=-1-2+4+8-16$, $9=-1-2+4-8+16$, $15=1+2+4-8+16$ and $-17=1+2+4-8-16$. Thus $f(n)=-1,-1,+1$ on the levels $0,1,2$ of the residues prime to 3, $f(n)=+1$ on the levels $0,1,2$ of the multiples of 3, and on the levels $3,4$ the digits depend on the unit part modulo 8 as displayed; see `digit_f.py`.

B Reproducibility

The folder `xxiv/` contains: `okada.py` (Okada's system), `pm1_search.py` (CP-SAT search), `digit_f.py` and `verify_E.py` (Theorem E and its checks), `exact_tree.py` and `star_search.py` (certificates for Table 2), `cert12.py` and `cert12_symbolic.py` (Theorem D), `class11.py` (class $(1,1)$), `structure_check.py` (Propositions 5.1 and 5.2 on data), `check_lemmas.py`, and the logs of all runs quoted above.

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